

Discrete random variable probability distributions



1

- a Yes, because $\sum P(x) = 1$, X is discrete numerical, and $0 \leq P(x) \leq 1$.
- b No, because even though X is discrete numerical, the other two conditions where $\sum P(x) = 1$ and $0 \leq P(x) \leq 1$ are not satisfied.

2

- a No b Yes c No d Yes

3

- a
- i
- $0 \leq P(x) \leq 1$
 - $\sum P(x) = 1$
 - X is discrete numerical.
- ii Yes
- b
- i
- $0 \leq P(x) \leq 1$
 - X is discrete numerical.
- ii No
- c
- i
- $0 \leq P(x) \leq 1$
 - $\sum P(x) = 1$
 - X is discrete numerical.
- ii Yes
- d
- i
- $0 \leq P(x) \leq 1$
 - X is discrete numerical.
- ii No

4

a

x	1	2	3
$P(X = x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$

- b Yes, because $\sum P(x) = 1$, X is discrete numerical, and $0 \leq P(x) \leq 1$.

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a

x	1	2	3
$P(X = x)$	$-\frac{1}{3}$	$-\frac{2}{3}$	$-\frac{3}{3}$

- b No, because even though X is discrete numerical, the other two conditions where $\sum P(x) = 1$ and $0 \leq P(x) \leq 1$ are not satisfied.

6

a

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$P(X = x)$	0	$\frac{1}{128}$	$\frac{1}{32}$	$\frac{9}{128}$	$\frac{1}{8}$

- b No, because even though X is discrete numerical and $0 \leq P(x) \leq 1$, the condition $\sum P(x) = 1$ is not satisfied.

7

a $\frac{1}{4}$
e $\frac{3}{16}$

b $\frac{7}{16}$
f $\frac{13}{16}$

c $\frac{7}{16}$

d $\frac{5}{16}$

8

a

x	0	1	2	3	4
$P(X \leq x)$	0.1	0.3	0.65	0.95	1
$P(X = x)$	0.1	0.2	0.35	0.3	0.05

b

i 0.65

ii $\frac{13}{14}$

9



a

x	$P(X \leq x)$	$P(X = x)$
0	0.001	0.001
1	0.011	0.010
2	0.055	0.044
3	0.172	0.117
4	0.377	0.205
5	0.623	0.246
6	0.828	0.205
7	0.945	0.117
8	0.989	0.044
9	0.999	0.01
10	1	0.001

b

i 0.172

ii 0.055

iii 0.066

iv 0.912

10

a $c = 60$ b $a = 2$ c $b = 3$

11

a $k = 0.2$ b $k = 0.3$

12

a $k = \frac{1}{11}$ b $\frac{4}{11}$ c $\frac{8}{11}$ d $\frac{4}{9}$ e $\frac{8}{9}$

13

a $k = \frac{20}{31}$ b $k = \frac{1}{50}$ c $k = 19$ d $k = \frac{576}{487}$

14

a

x	0	1	2	3
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

b

x	0	1	2	3
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{5}{12}$

c

x	0	1	2	3
$P(X = x)$	$\frac{1}{13}$	$\frac{2}{13}$	$\frac{2}{13}$	$\frac{8}{13}$



d

x	0	1	2	3
$P(X = x)$	$\frac{1}{14}$	$\frac{5}{14}$	$\frac{5}{14}$	$\frac{3}{14}$

15

a $a = 0.23$

b $b = 0.16$

16

a $c = 0.22$

b $a = 0.24$

c $b = 0.21$

17

a $k = \frac{1}{15}$

b

x	4	5	6	7	8
$P(X = x)$	$\frac{5}{15}$	$\frac{4}{15}$	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$

c

$\frac{4}{5}$

d

$\frac{2}{5}$

e

$\frac{2}{3}$

f

$\frac{9}{14}$

18

a

x	0	1	2	3	4	5
$P(X = x)$	0.01024	0.07680	0.23040	0.34560	0.25920	0.07776

b Negatively skewed

c 0.33696

d 0.98976

e 0.65955

19

a Geometric

b

$\frac{1}{2}$

c

$\frac{1}{2}$

d 1

e

$\frac{7}{32}$

f

$\frac{508}{511}$



Uniform distribution

20

a

x	0	2	4	6	8	10
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b Yes, because $\sum P(x) = 1$, X is discrete numerical, and $0 \leq P(x) \leq 1$.

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a

$$k = \frac{1}{6}$$

b

$$P(X = x) = \begin{cases} \frac{1}{6}; & x = 1, 2, 3, 4, 5, 6 \\ 0, & \text{otherwise} \end{cases}$$

22

a

$$k = \frac{1}{5}$$

b $\frac{2}{5}$

c $\frac{4}{5}$

d $\frac{3}{4}$

e

$$\frac{3}{4}$$

23

a

$$\frac{1}{4}$$

b $\frac{1}{2}$

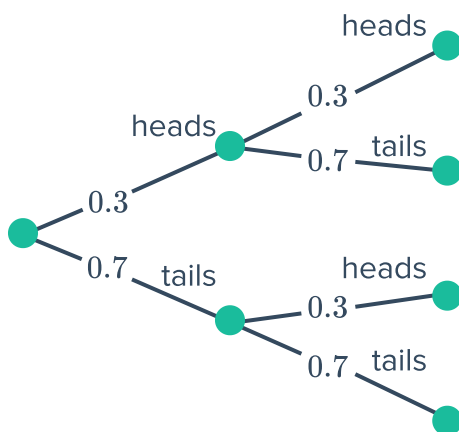
c $\frac{2}{3}$

d 2

Applications

24

a



b

x	0	1	2
$P(X = x)$	0.09	0.42	0.49

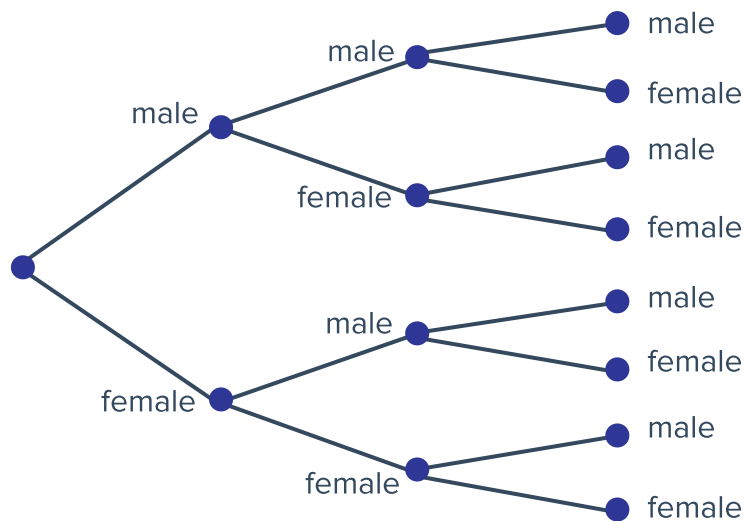
- c
- X is discrete numerical
 - $0 \leq P(x) \leq 1$
 - $\sum P(x) = 1$



d Yes

25

a



b

f	0	1	2	3
$P(F = f)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

c Non-uniform

26

a 1, 2, 3, 4, 5, 6

b Uniform, the outcomes are equally likely.

c

x	1	2	3	4	5	6
$p(x)$	$\frac{996}{6000}$	$\frac{1005}{6000}$	$\frac{994}{6000}$	$\frac{1005}{6000}$	$\frac{996}{6000}$	$\frac{1004}{6000}$

d $\frac{667}{2000}$

e 0.05%

27

a

i $\frac{1}{8}$

ii $\frac{49}{512}$

iii $\frac{343}{4096}$

iv $\frac{117\,649}{2\,097\,152}$

- b
- X is discrete numerical.
 - $\sum P(x) = 1$
 - $0 \leq P(x) \leq 1$



c Yes

28

a

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

b $\frac{1}{3}$

c $\frac{1}{5}$

29

a $k = \frac{1}{21}$

b

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

30

a 0.1251

b Yes, the distribution is approximately symmetrical with the mean, median and mode coinciding at 10 Jupiter bars.

c

i 0.0948

ii 0.0901

iii 0.0103

iv 0.1198

d Yes the claim is correct as $P(8 \leq X \leq 12) = 0.5713$, which is higher than 50%.

31

a

x	38	39	40	41	42
$P(X = x)$	$\frac{5}{70}$	$\frac{7}{70}$	$\frac{24}{70}$	$\frac{8}{70}$	$\frac{26}{70}$

b

i $\frac{29}{35}$

ii $\frac{18}{35}$



33

a $k = 0.8$

b 0.8

c 0.000256